

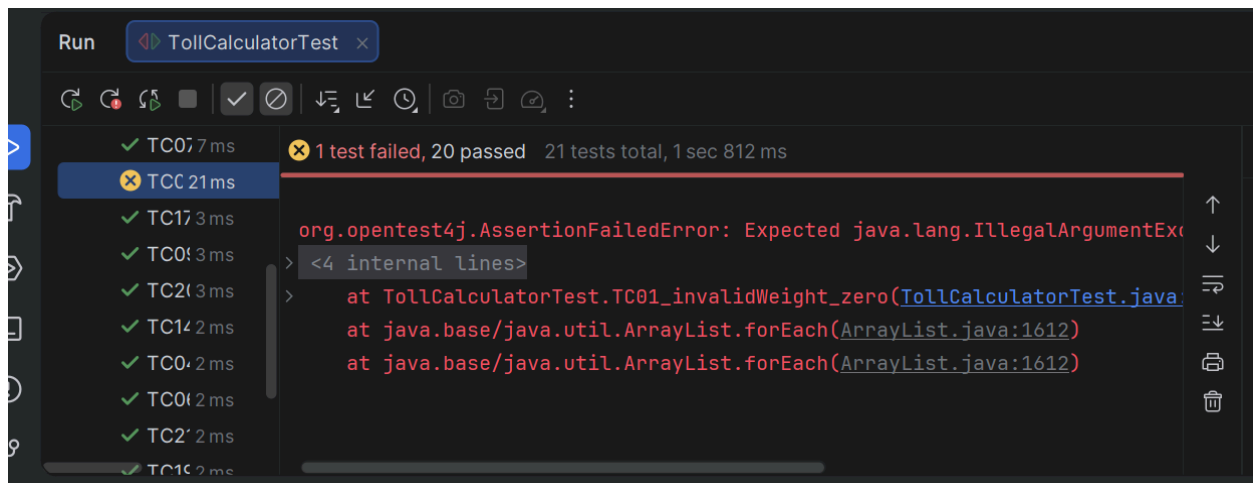
Github Repo: https://github.com/MuhammadBasit29/MuhammadBasit_ST_Exam1

Fault1

Changed weight ≤ 0 to weight < 0 , so EP catches the invalid weight = 0 case.

```
//      if (weight <= 0) {  
//          throw new IllegalArgumentException("Weight must be positive.");  
//      }  
//      Fault 1  
      if (weight < 0) {  
          throw new IllegalArgumentException("Weight must be positive.");  
      }
```

Result



Fault 2

Changed weight > 1200 to weight ≥ 1200 , so BVA catches the boundary error at weight = 1200.

```

//      if (weight > 1200 && weight <= 3500) {
//      Fault 2
//      if (weight >= 1200 && weight <= 3500) {
//          if (isEV && isCarpool) {
//              discountPercent = 0.15;
//          } else if (isEV || isCarpool) {
//              discountPercent = 0.10;
//          }
//      }

```

Result

Run **TollCalculatorTest** x

1 test failed, 20 passed 21 tests total, 1 sec 838 ms

org.opentest4j.AssertionFailedError:
Expected :0.0
Actual :0.15
[Click to see difference](#)

> <6 internal lines>
> at TollCalculatorTest.TC09_weight1200(TollCalculatorTest.java:96) <1 :
> <2 folded frames>

Performance

Show Results

CPU

Heap Memory

Fault 3

Changed Tier-2 isEV

```

    }
    //      else if (isEV || isCarpool) {
    //          discountPercent = 0.10;
    //      }
    //      FAULT 3
    //      else if (isEV && isCarpool) {
    //          discountPercent = 0.10;
    //      }
    } else if (weight > 3500) {

```

Result

Run TollCalculatorTest x

2 tests failed, 19 passed 21 tests total, 1 sec 297 ms

org.opentest4j.AssertionFailedError:
Expected :0.1
Actual :0.0
[<Click to see difference>](#)

> <6 internal lines>
> at TollCalculatorTest.TC12_tier2_EVOnly([TollCalculatorTest.java:127](#))
> <2 folded frames>

- TC1 13 ms
- TC0! 2 ms
- TC0! 2 ms
- TC12 7 ms
- TC16 3 ms
- TC0! 2 ms
- TC0! 3 ms
- TC17 2 ms
- TC0! 2 ms
- TC2 1 ms



Fault 4

Code Coverage: A hypothetical statement was added for weight == 5000. None of the existing EP, BVA, or Decision Table test cases uses a weight of 5000, so the statement is not executed. Code Coverage identifies the statement as uncovered.



Fault 5

The Tier 3 condition `isEV && isCarpool` is hypothetically changed to `isEV || isCarpool`. This changes the decision branches for EV-only and Carpool-only vehicles. Branch Coverage is used to ensure the alternative outcomes of the decision are exercised, exposing the incorrect branch behavior.

```
// FAULT 5
if (isEV || isCarpool) {
    discountPercent = 0.25;
}

//      if (isEV && isCarpool) {
//          discountPercent = 0.25;
//      }
else {
```

Result

✖ 2 tests failed, 19 passed 21 tests total, 1 sec 238 ms

org.opentest4j.AssertionFailedError:

Expected :0.05

Actual :0.25

[<Click to see difference>](#)

> <6 internal lines>

> at TollCalculatorTest.TC19_tier3_EVOnly(TollCalculatorTest.java:200)

> <2 folded frames>

✖ 2 tests failed, 19 passed 21 tests total, 1 sec 238 ms

org.opentest4j.AssertionFailedError:

Expected :0.05

Actual :0.25

[<Click to see difference>](#)

> <6 internal lines>

> at TollCalculatorTest.TC20_tier3_carpoolOnly(TollCalculatorTest.java:?)

> <2 folded frames>